INDRAJAL : The Firefighting Aircraft

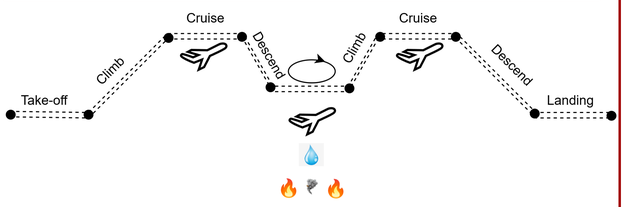
Department of Mechanical & Aerospace Engineering

IOE Pulchowk Campus

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Mission Profile



Design Requirements

Utility Aircraft: Firefighting Aircraft
Short Take off and Landing
Carries: Pilot and Copilot
Cruise Ceiling: 4800 m
STOL aircraft
Cruise speed: 75 m/s

Aircraft Specification

General Specification	
Engine Power	2*650 hp
Range	500 km
Endurance	2 hr
No. of crew	2

Weights	
MTOW	8400 kg
Empty Weight	3600 kg
Fuel Capacity	1300 kg
Payload Weight	3500 kg

Wings	
Configuration	High wing
Airfoil	NACA 2412
Span	20 m
MAC	2.56 m
Aspect Ratio	7.81
Projected Area	51.24 m^2
Dihedral Angle	2 deg

Performance	
Take off distance	426 m
Landing distance	518 m
Cruise speed	75 m/s
Cruise Altitude	4800 m
Rate of Climb	7 m/s
Cruise Ceiling	6500 m

Horizontal Stabilizer	
Airfoil	NACA 0008
Span	6m
Aspect Ratio	2.61

Vertical Stabilizer	
Airfoil	NACA 0008
Span	4m
MAC	2.45 m
Taper Ratio	0.47

Loading & Droping System

- Water Fill rate about 1135 liters per minute
- 63.5mm hose diameter GEN II Fire Retardant Drop System (FRDS)
- Max. drop of 3500 liters in 20 seconds
- Garmin G1000 NXi: Integrated Glass System Cockpit





Introduction

A twin-engine STOL (Short Takeoff and Landing) utility aircraft specially designed to assist in controlling and suppressing wildfires by delivering water directly onto the burning areas.

- Equipped with pressurized drop system
- Uses compressed air to push water/retardant through belly-mounted nozzles
- Operation at diverse altitudes and challenging conditions.
- Two-people crew: Pilot and Co-pilot

AIRCRAFT LAYOUT

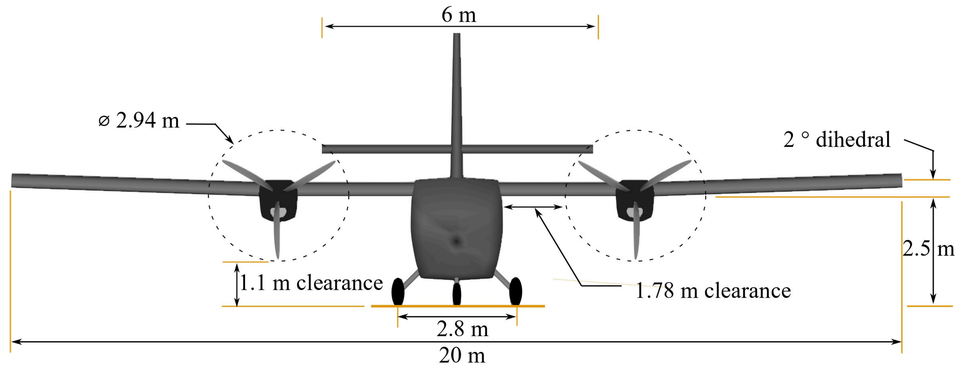


Fig: Front View

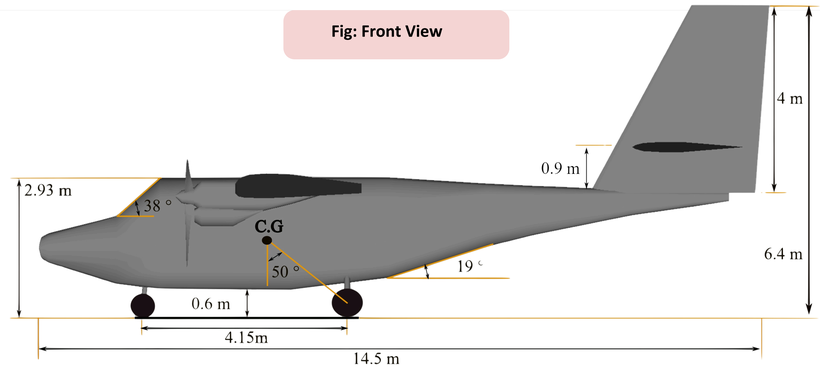
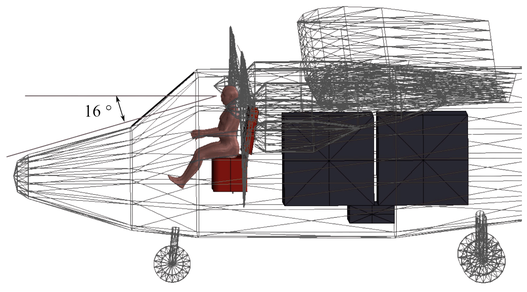
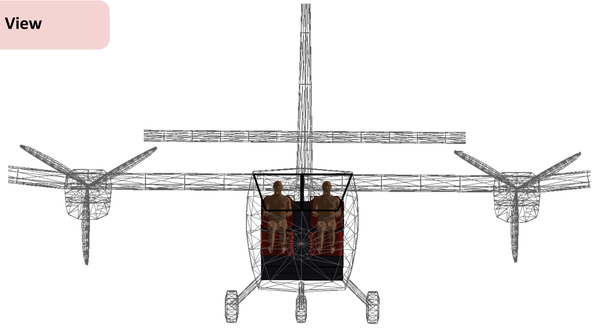


Fig: Side View

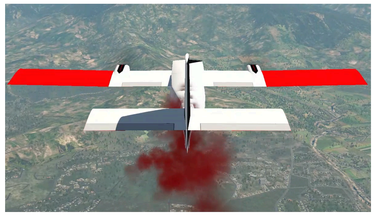




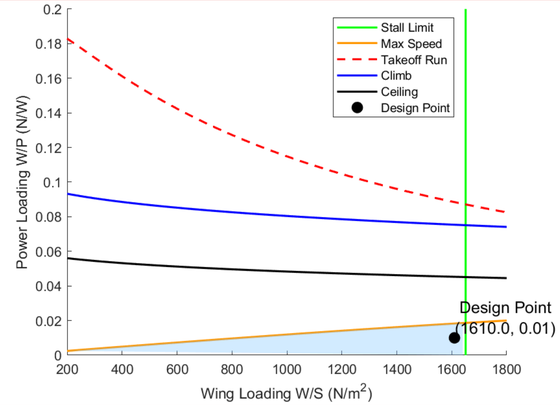
Water Drop Test



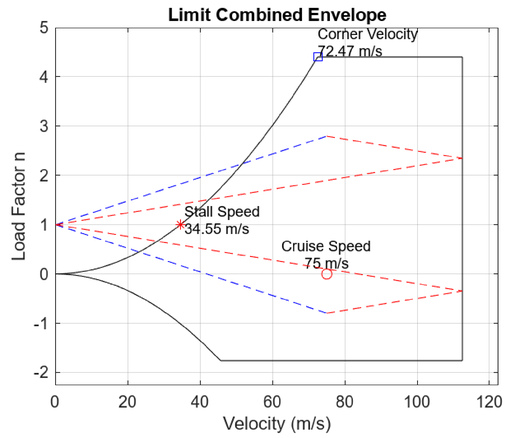
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Matching Plot



Flight Envelope and Gust Envelope



Static Stability

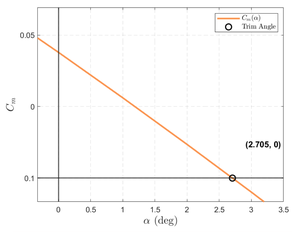


Fig: Cm vs alpha

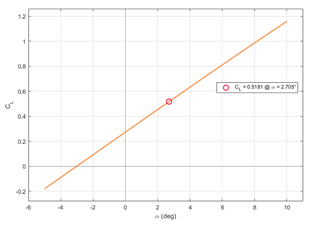


Fig: Lift Curve

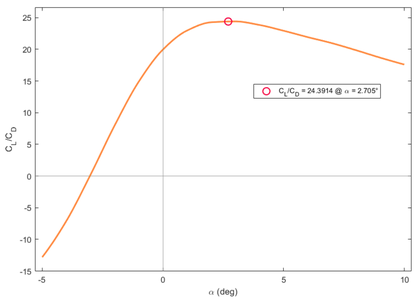


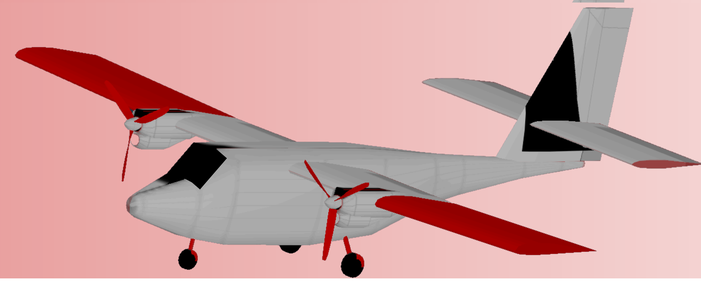
Fig: Lift to Drag Ratio

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Dynamic Stability Test

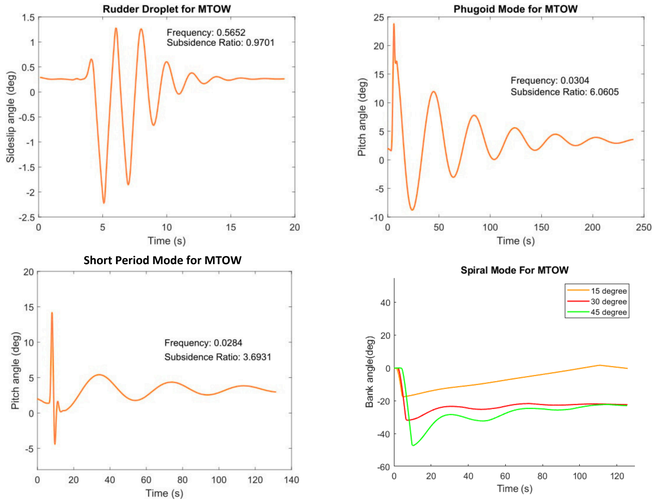


Fig: Dynamic Characteristics

Neutral Point Determination

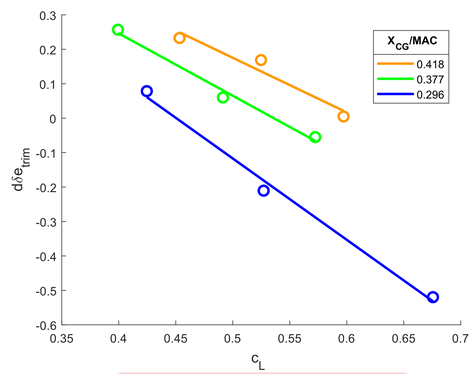


Fig: Elevator trim Vs Cl

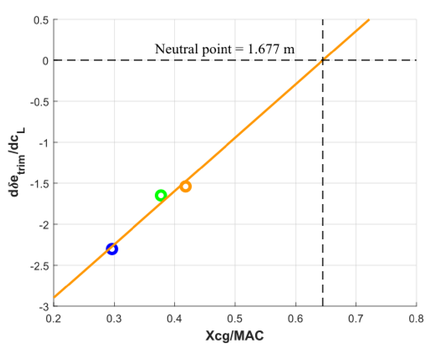


Fig: Determination of Neutral Point

CG Envelope

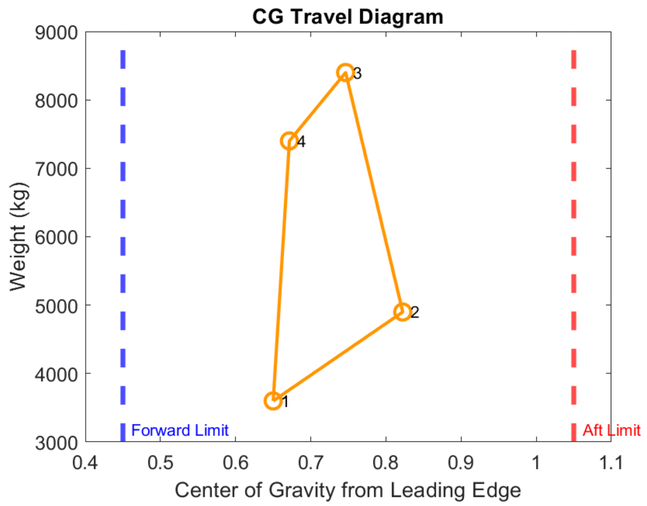


Fig: CG Travel Diagram

Handling Quality Test

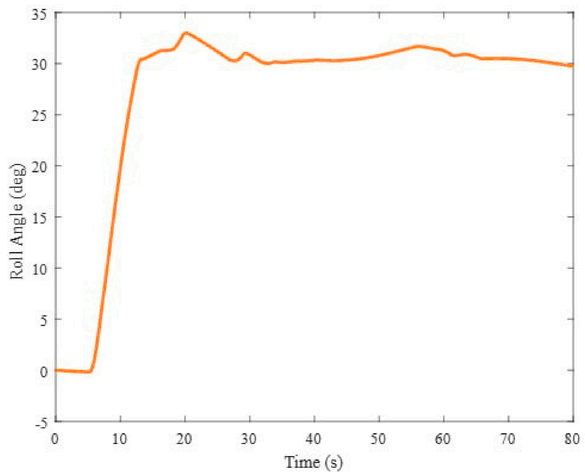


Fig : Bank Angle vs Time

The aircraft was rated '3' as per Cooper Harper Rating Scale. The aircraft had some mild unpleasant deficiencies and minimal pilot compensation was required for desired characteristics.

Conclusion

Indrajal successfully completed its flight tests, proving its strong performance and safe handling in firefighting missions. The aircraft showed dependable stability and control, with the ability to operate from short or rough runways. With a range of 500 km and endurance of 2 hours, it can carry around 3,500 liters of water or retardant, making it well-suited for quick and effective wildfire response.

Stall Characteristics

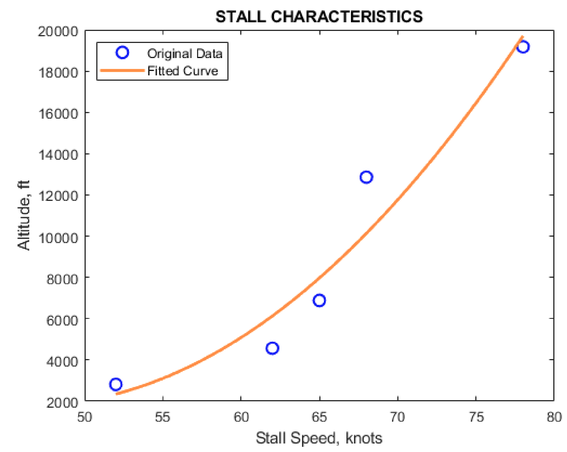


Fig : Altitude vs Stall Speed

Climb and Descent Characteristics

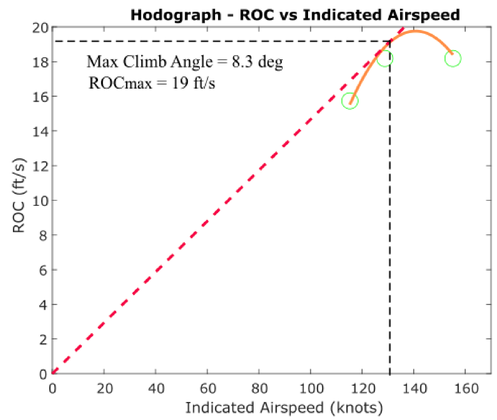


Fig: ROC vs Indicated Airspeed

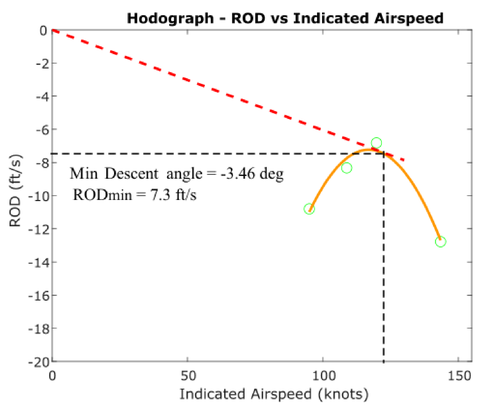


Fig: ROD vs Indicated Airspeed